

December 11th, 2025

Dear Editor,

We are submitting our manuscript, “High temperatures predict altered gut microbiome composition, metabolic capacity, and immune interactions in a wild primate species,” to *Proceedings B*. Here, we determine the influence of natural thermal variation on gut microbiome composition and metabolic capacity in a wild population of capuchin monkeys (*Cebus imitator*). We find that high temperatures are associated with altered community-level diversity, taxonomic composition, and metabolic capacity consistent with physiological responses to heat stress. We also show that the microbiome’s response to high temperatures is linked to host immunity and may affect gut-immune interactions. Previous experimental work has demonstrated that microbiomes play a key role in determining the thermal tolerances of their host species, but this research has largely been conducted in laboratory settings on endotherms and livestock, and offers limited insight into how species respond to natural thermal variation. Our results provide rare evidence from a natural population of primates that changes to the gut microbiome at high temperatures may contribute to physiological heat stress by compromising essential microbiome functions. Our results also highlight an important pathway by which rising temperatures due to climate change may threaten wild animal populations, particularly in tropical species with narrow thermal tolerances.

This manuscript describes original work and is not under consideration by any other journal. All authors approved the manuscript and this submission. All protocols involved in this study were previously approved by the University of Michigan IACUC (#PRO00011898) and comply with the relevant legal requirements governing animal research in the United States of America and Costa Rica.

Thank you for considering our manuscript. We appreciate your time and look forward to your response.

Sincerely,

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